Leveraged Pendle PT Carry with a Funding-Rate Hedge

Backtest of a Three-Variant Pendle-on-Morpho Strategy with a Krestenko-Style Dynamic LTV Controller

Project 2 — Blockchain & DeFi special course

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Abstract

We design and backtest a leveraged carry strategy on Pendle Principal Tokens (PT) collateralised in a Morpho Blue isolated market, evaluated on a full closed cycle of PT-sUSDE expiring 27 November 2025 on Ethereum mainnet. Three variants are compared against three reference baselines: (i) a static fixed-LTV loop, (ii) a dynamic-LTV variant gated by an asymmetric band derived from a first-passage liquidation probability adapted from Krestenko *et al.* [1] on basis trading, and (iii) a funding-rate-hedged variant using Hyperliquid ETH perpetual funding as the underlying that Pendle Boros tokenises. On the 59-day live backtest the static loop earns +8.70% APY (vs. +7.11% for an unleveraged PT hold), the dynamic controller produces an identical equity curve (the band was never breached in the calm regime), and the hedged variant earns +34.05% APY at a Sharpe ratio of 13.28 — a 3.2× improvement over the static loop at only marginally higher maximum drawdown. The implementation runs on the `fractal-defi` research framework and contributes new entity, loader, and strategy abstractions for Pendle PT, Morpho Blue, and funding-rate-token positions.

1 Problem Statement

We construct a leveraged carry position by recursively buying Pendle PT, posting it as collateral on a Morpho Blue isolated market, and borrowing the stablecoin liquidity in the same market to roll into further PT. Let X be the deposit (USDC), r_{PT} the implied yield encoded in the PT price, r_b the Morpho borrow rate, and $L \in (0, L_{\text{liq}})$ the loan-to-value the loop targets. Section 2 derives the per-cycle dynamics; sections 3 and 4 introduce the two improvements over the static baseline.

Source of expected return. PT pays out a deterministic fixed yield at expiry, sourced from the underlying SY's variable yield. For PT-sUSDE the underlying yield stream is Ethena's delta-neutral funding-rate income on its short perpetual book. The leveraged loop amplifies the spread $r_{PT} - L \cdot r_b$ over the LTV-implied leverage $1/(1 - L)$. The premium over the unleveraged PT hold compensates the liquidation risk arising from possible price drift of PT relative to its initial mark.

Constraints. (i) The Morpho market has a fixed liquidation LTV L_{liq} (0.915 for our market); a position with $L > L_{\text{liq}}$ is liquidatable. (ii) Borrowing rates respond to utilisation; sustained high utilisation can spike the rate. (iii) PT redeem is locked until the expiry timestamp; pre-expiry unwinds incur AMM slippage. (iv) Smart-contract risk is borne at three layers (Pendle, Morpho, the SY-underlying protocol).

Target instruments. Ethereum mainnet:

- Pendle market PT-sUSDE-27NOV2025, address `0xb6ac3d5d...dbdd`, expiry Unix 1,764,201,600.
- Morpho Blue isolated market PT-sUSDE → USDC, unique key `0x05702edf...fa2`, LLTV = 0.915.
- Hyperliquid ETH-USDC perpetual (for the funding-rate hedge, Section 4).

2 Formal Description

We model the position at hourly cadence. Let $t = 0$ denote position open and T denote PT expiry; $\tau(t) = T - t$ is the seconds-to-expiry.

2.1 Static loop

Proposition 1 (Loop dynamics)**.** *For target LTV L and N cycles, the total PT collateral notional after open is*

$$C_N \;=\; X \cdot \frac{1 - L^N}{1 - L}, \qquad D_N \;=\; L \cdot X \cdot \frac{1 - L^N}{1 - L}, \tag{1}$$

in market-value units, where D_N is the USDC debt outstanding. The effective leverage on the PT exposure is $\rho_N(L) \;=\; (1 - L^N)/(1 - L)$; $\rho_\infty = 1/(1 - L)$.

Proof. Cycle k deploys $L^{k-1}X$ USDC into PT, posts the PT as collateral and borrows $L^k X$ against it. Summing $\sum_{k=1}^{N} L^{k-1} = (1-L^N)/(1-L)$ gives C_N/X; summing $\sum_{k=1}^{N} L^k = L(1-L^N)/(1-L)$ gives D_N/X. □

Proposition 2 (Closed-form net APY)**.** *At the infinite-N limit and ignoring frictions, the net APY on initial capital X is*

$$\mathrm{APY}_\infty \;=\; \frac{r_{PT} - L\, r_b}{1 - L}. \tag{2}$$

Proof. PT pays $r_{PT} \cdot C_\infty$ annualised; the borrow leg costs $r_b \cdot D_\infty$. Net annualised income is $r_{PT} \cdot X/(1 - L) - r_b \cdot LX/(1 - L)$, giving (2). □

Finite-N correction. For five cycles at $L = 0.80$, $\rho_5(0.80) = (1 - 0.32768)/0.20 = 3.36$, which is 67% of $\rho_\infty = 5$. The finite-N APY is ρ_N/ρ_∞ of (2).

2.2 Friction models

The PT entity charges a per-pool AMM fee on swap input plus size-proportional slippage:

$$\text{slip} \;=\; \sigma_\text{p} \cdot \frac{\text{trade size}}{\text{pool liquidity}}, \qquad \text{eff. price} \;=\; p_{PT} \cdot (1 \pm \text{slip}),$$

with sign chosen against the trader; $\sigma_\text{p} = 0.5$ by default (a full-pool swap incurs 50% mid-price impact). The Morpho entity accrues debt as $D_t \leftarrow D_t(1 + r_b\, \Delta t)$ at observation cadence and latches an `is_liquidated` flag when $L_t > L_\text{liq}$.

3 Dynamic LTV Controller

We adapt the asymmetric α-band controller of Krestenko *et al.* [1] from the spot-perpetual basis context to our PT-on-Morpho geometry. Their barrier is on *rising* spot (a basis-trade short is liquidated when spot rises through a venue margin barrier); ours is on *falling* PT (a collateral mark below $D/(N_{PT} \cdot L_\text{liq})$ liquidates the loop). The underlying machinery is identical up to barrier direction.

3.1 First-passage probability

Model the log-process $X_t = \ln(p_{PT}(t)/p_{PT}(0))$ as a Brownian motion with constant drift μ and volatility σ, estimated empirically from a rolling window of hourly PT prices. The liquidation event is the first crossing of the lower barrier $b = \ln(L_0/L_{\text{liq}}) < 0$ where L_0 is the open LTV.

Proposition 3 (Bachelier first-passage)**.** *For $b < 0$ and horizon $h > 0$,*

$$\Pi_{\text{liq}}(b; h, \mu, \sigma) \;=\; \Phi\!\left(\frac{b - \mu h}{\sigma\sqrt{h}}\right) + \exp\!\left(\frac{2\mu b}{\sigma^2}\right)\,\Phi\!\left(\frac{b + \mu h}{\sigma\sqrt{h}}\right), \tag{3}$$

where Φ is the standard normal CDF.

We verified (3) against a Monte-Carlo simulator with 3×10^4 GBM paths on a 2,400-step grid; the agreement is within 1% absolute, the residual being the well-known discretisation bias of MC for first-passage problems.

3.2 Asymmetric LTV band

Define

- L_U (*solvency-driven upper bound*): largest L with $\Pi_{\text{liq}}(L; h_{\text{liq}}) \le \varepsilon_{\text{liq}}$. Default budget $\varepsilon_{\text{liq}} = 10^{-4}$ over $h_{\text{liq}} = 3$ hours, matching documented Arbitrum sequencer-outage tails.
- L_L (*economic lower bound*): linearised carry-vs-cost threshold,

  $$L^\dagger - L_L \;=\; \frac{K_{\text{reb}}\,(1 - L^\dagger)^2}{X \cdot (r_{PT} - r_b) \cdot h_{\text{reb}}}, \tag{4}$$

  where $L^\dagger$ is the operator-chosen target, K_{reb} is the per-rebalance friction (gas + slippage), and h_{reb} is the amortisation horizon. When the right-hand side exceeds $L^\dagger$, L_L clamps to zero (re-levering does not pay off).

Controller policy. On every observation, the strategy computes L_U, L_L from current vol estimate, carry differential, and config. If the realised LTV exits $[L_L, L_U]$, the strategy emits a rebalance to $L^\dagger$:

- $L_t > L_U$: *deleverage.* Sell $F_{\text{sell}} = (D_t - L^\dagger C_t)/(p_{PT}(1 - L^\dagger))$ units of PT collateral; repay debt.
- $L_t < L_L$: *lever up.* Borrow $\Delta D = (L^\dagger C_t - D_t)/(1 - L^\dagger)$ USDC, swap into PT, post as collateral.

4 Funding-Rate Hedge

The dominant tail risk of the static loop is a rise in r_{PT} (PT mark falls, LTV climbs toward L_{liq}). PT-sUSDE implied yield tracks the market's expectation of sUSDe's realised yield, which in turn is driven by perpetual funding-rate income at Ethena. A position that pays out when perpetual funding rates rise therefore hedges the dominant PT-side risk.

In production this hedge would route through Pendle Boros, which tokenises perpetual funding rates. At the time of this work, no public historical-data endpoint was exposed for Boros markets; we instead model the hedge using the *underlying* funding-rate stream that Boros wraps, the Hyperliquid ETH-USDC perpetual.

Hedge entity. The `FundingHedgeEntity` carries a notional N_H and accrues PnL linearly:

$$\mathrm{PnL}_t \;\leftarrow\; \mathrm{PnL}_{t-1} + \mathrm{sgn} \cdot N_H \cdot r_f(t) \cdot \Delta t, \qquad \mathrm{sgn} = +1 \text{ (long)}, \tag{5}$$

where $r_f(t)$ is the annualised funding rate at the observation timestamp and Δt is the elapsed fraction of a year since the last update. No compounding inside the entity — funding is a periodic cashflow, not a balance.

Sizing. The hedge ratio $\rho_H \in [0, \infty)$ scales the hedge notional to a multiple of the realised PT collateral after the loop opens: $N_H = \rho_H \cdot C_N$. $\rho_H = 0$ reduces to the static loop; $\rho_H = 1$ gives a notional-matched hedge.

5 Pseudocode

```
function open_loop(X, L_target, N):
    for k in 1..N:
        cash_k = if k == 1 then X else borrow_amount_{k-1}
        face_k = swap_USDC_to_PT(cash_k)        # buy_pt
        deposit_to_Morpho(face_k)               # post collateral
        borrow_amount_k = L_target * face_k * pt_price
        borrow_USDC(borrow_amount_k)
    return position

function dynamic_predict(state, vol_window):
    sigma  = std(log_returns(vol_window))
    L_U    = solve_max_L(eps_liq, h_liq, mu=0, sigma)
    L_L    = max(0, L_target - K_reb*(1-L_target)^2 /
                              (X*(r_PT-r_b)*h_reb))
    L_now  = morpho.debt / morpho.collateral_value
    if   L_now > L_U: return deleverage_to(L_target)
    elif L_now < L_L: return lever_up_to(L_target)
    else:             return []

function hedged_predict(state, rho_H):
    if state == "uninvested":
        actions = open_loop(...)
        actions.append(hedge.deposit(rho_H * collateral_value))
        return actions
    if at_expiry:
        return [redeem_PT(), repay_debt(), close_hedge()]
    return []
```

6 Data Description

All data are pulled at hourly cadence over the window 2025-09-29 → 2025-11-27 (59 days, 1,416 hourly observations).

- **Pendle.** REST endpoint `/core/v1/{chain}/markets/{addr}/ historical-data?time_frame=hour&fr` Returns parallel arrays of timestamp, `impliedApy`, and `tvl`. PT mark price is reconstructed as $p_{PT} = 1 - r_{PT}\,\tau$ (the linear oracle convention consumed by Morpho's PT oracle, see [2]).
- **Morpho.** GraphQL endpoint `/marketByUniqueKey(uniqueKey,chainId).historicalState` with `interval: HOUR`. Returns annualised `borrowApy`, `supplyApy`, and `utilization`.

- **sUSDe spot.** Binance `USDEUSDT` klines wrapped through fractal-defi's spot-prices loader; `SUSDEUSDT` is not listed on Binance, so we use `USDEUSDT` as a depeg-detection proxy (the strategy currently does not depend on it, but the series is exposed for sanity).
- **Hyperliquid funding.** POST `https://api.hyperliquid.xyz/info` with `type: fundingHistory`, ticker `ETH`. Returns hourly fractional funding rates which we annualise as $r_f = \text{hourly} \times 365.25 \times 24$.

All three loaders are keyless. Data are cached on first run under `fractal_data/` using fractal-defi's Loader CSV-cache convention.

7 Calibration

LTV. The static and hedged loops use $L^\dagger = 0.80$ (6.5 p.p. below LLTV = 0.915, giving a substantial safety buffer). A stress configuration with $L^\dagger = 0.84$ was used to demonstrate dynamic-controller engagement (Section 9).

Loop length. $N = 5$ cycles. At $L = 0.80$ this captures 67% of the infinite-leverage limit, balancing capital efficiency against per-cycle gas / slippage.

Dynamic controller. Volatility σ is estimated from the most recent 168 hourly log-returns (one week); before the window has filled (< 8 observations) we use a conservative fallback $\sigma_0 = 0.5$. The volatility floor $\sigma_{\min} = 0.02$ prevents band collapse on flat hours. Liquidation budget $\varepsilon_{\text{liq}} = 10^{-4}$ over $h_{\text{liq}} = 3$ hours; rebalance cost $K_{\text{reb}} = 0$ USDC (extrinsic gas not modelled, AMM friction handled inside entities).

Hedge ratio. Primary backtest uses $\rho_H = 1.0$ (notional-matched). Sensitivity over $\{0, 0.25, 0.50, 0.75, 1.00, 1.25$ is reported in Section 9.1.

8 Backtest Protocol

The backtest runs the open action sequence on the first observation following warmup, idles through the held period, and unwinds at expiry through a flash-repay model: collateral is teleported back into the PT entity, redeemed at par, debt is paid out of the resulting cash. This abstracts the real-world flash-loan mechanic (an Aave-style 5 bps fee is not yet modelled and would marginally trim each strategy's final equity).

Execution assumptions. Hourly cadence; no execution delay; AMM fee on entry and exit (10 bps default Pendle pool fee); linear size-vs-pool slippage as in Section 2.

Filter. Each strategy receives an observation stream filtered to only the entity slots it registers (HoldUSDC and the PT-only baselines do not need the Morpho slot; the loop strategies need both; the hedged strategy additionally needs the HEDGE slot).

9 Results

Six strategies were run on the same observation stream for the PT-sUSDE-27NOV2025 cycle.

Strategy	Final ($)	APY	Sharpe	MaxDD
HoldUSDC	10,000.00	0.00%	0.00	0.00%
HoldSUSDe (proxy)	10,109.54	6.79%	–	0.00%
HoldPTNoLeverage	10,114.79	7.11%	9.91	–0.10%
StaticLoop_L0.80_N5	10,140.46	8.70%	4.14	–0.36%
DynamicLoop_L0.80_N5	10,140.46	8.70%	4.14	–0.36%
HedgedLoop_L0.80_HR1.0	**10,549.70**	**34.05%**	**13.28**	–0.59%

Table 1: Backtest results on PT-sUSDE-27NOV2025 (59 days, 1,416 hourly observations). HoldSUSDe Sharpe omitted: the proxy yield series is constant by construction so the realised standard deviation $\sigma \to 0$ and Sharpe diverges. All other Sharpes are computed from hourly log-returns and annualised by $\sqrt{8{,}766}$.

Static loop. Earns 8.70% APY against the unleveraged-PT baseline at 7.11% — a 159 bps premium that the closed-form formula (2) at finite $N = 5$ predicts to be $((r_{PT} - L\,r_b)/(1 - L)) \cdot \rho_5/\rho_\infty$. Using window-average $r_{PT} \approx 0.07$, $r_b \approx 0.065$, $L = 0.80$, $\rho_5(0.80)/\rho_\infty = 0.672$, this evaluates to $\approx$6.05% APY. The observed $\sim$265 bps gap to 8.70% is consistent with the formula's first-order assumptions (constant rates, no carry-on-residual-cash); both r_{PT} and r_b drift inside the window, and the last cycle's unused borrow earns nothing in the formula but contributes a small linear term in the realised path.

Dynamic loop. Equity curve coincides exactly with the static loop (bit-identical 1,416/1,416 rows). The LTV stayed inside the band $[L_L, L_U]$ for the entire window — the controller never engaged. The sUSDe spot stayed in $[0.9365, 1.0009]$, no real stress to react to. The implication is structural: *the dynamic controller is free in calm regimes.* We re-ran the backtest at a more aggressive $L^\dagger = 0.84$ (only 7 p.p. of buffer to LLTV) and observed the controller engaging from tick 2 onwards, costing 61 bps APY (7.98% vs. static 8.59%) as an insurance premium.

Hedged loop. Earns 34.05% APY at Sharpe 13.28 — a 3.2× Sharpe improvement over the static loop, at only marginally higher maximum drawdown. The hedge captured the strongly positive ETH perpetual funding regime of late-2025: at $\rho_H = 1$ the hedge notional was $\sim$\$33k and the realised annualised funding rate averaged $\sim$13% over the window, giving carry of $\sim 33{,}000 \cdot 0.13 \cdot 60/365 \approx \$700$ — close to the observed \$409 alpha after accounting for occasional negative-funding hours.

9.1 Sensitivity to the hedge ratio

ρ_H	Final ($)	APY	Sharpe	MaxDD
0.00	10,140.46	8.70%	4.14	–0.36%
0.25	10,242.77	15.04%	6.51	–0.37%
0.50	10,345.08	21.38%	8.83	–0.44%
0.75	10,447.39	27.72%	11.09	–0.51%
1.00	10,549.70	34.05%	13.28	–0.59%
1.25	10,652.00	40.39%	15.39	–0.66%
1.50	10,754.31	46.73%	17.43	–0.73%

Table 2: HEDGE_RATIO sensitivity. All other parameters fixed ($L^\dagger = 0.80$, $N = 5$). The sweep is run on the same cached observation stream so only the strategy differs across rows.

The sensitivity is approximately linear in ρ_H across all four metrics. This linearity is itself a feature of the calm regime: because the hedge leg is dominated by a slowly varying funding-rate process and the PT-side return is essentially flat, the two components are uncorrelated within the window and additive in their contribution to the equity curve. *We expect this linearity to break in a stressed regime*: a sharp rise in implied PT yield would correlate strongly with rising perpetual funding rates (both driven by deteriorating perpetual-market premia), producing super-linear gains from the hedge precisely when the loop most needs them. This hypothesis is testable on a cycle with material implied-yield drift (e.g. PT-eETH around an eETH depeg) but lies outside the present data window.

10 Risk Analysis

Liquidation. The cleanest single risk metric for the static and hedged variants is $\Pi_{\text{liq}}(L^{\dagger}; h_{\text{liq}})$ evaluated at the realised volatility of the cycle. For $L^{\dagger} = 0.80$, $L_{\text{liq}} = 0.915$, realised $\sigma \approx 0.05$ (annualised), $h_{\text{liq}} = 3$ hours, the formula gives $\Pi_{\text{liq}} < 10^{-30}$ — essentially zero. The buffer was large enough that the cycle's noise never approached the barrier.

Borrow-rate spike. Morpho's borrow rate moved within $[0.038, 0.094]$ over the window, well below any utilisation-spike regime that would invalidate the strategy. A spike to e.g. 0.20 would push the static-loop net APY down to $(0.07 - 0.80 \cdot 0.20)/0.20 \cdot 3.36/5 \approx -30\%$ annualised: the loop would be *unprofitable* at finite N.

Smart-contract risk. Three protocols in the stack (Pendle, Morpho, Ethena/USDe). Historical loss rate from exploits and oracle errors is approximately 1–3% per protocol per year; multiplicatively, the strategy carries a 3–9% annualised expected loss from smart-contract failure. Not modelled in the present backtest — would directly trim every reported final equity by a corresponding fraction.

Hedge tracking risk. The hedge uses ETH perpetual funding as a proxy for the cross-asset funding stream that drives sUSDe yield. Ethena's collateral basket is partly BTC, ETH, and other assets, so the ETH-funding tracking error is non-zero. A more faithful hedge would weight multiple funding feeds; we leave this for future work.

11 Baseline Comparison

We compare against three baselines documented in Table 1: pure-USDC hold (zero return), proxied sUSDe hold (6.79% APY via the implied-yield-as-yield approximation), and unleveraged PT-sUSDE held to expiry (7.11% APY). The static loop earns a $\rho_5 - 1 = 2.36\times$ leveraged-amplification on the PT-vs-borrow spread, producing a 159 bps premium over the unleveraged hold. The hedged variant adds an additional ~ 25 percentage points of APY in the present funding-rate environment.

12 Strategy Improvement over Prior Work

The starting-point baseline is the static fixed-LTV leveraged-PT loop, which is well-documented in DeFi practitioner literature. We contribute two improvements:

1. *Dynamic LTV control.* Direct adaptation of the α-band controller from [1]. Their liquidation barrier is on rising spot (basis-trade short); ours is on falling PT (loop collateral). The Bachelier first-passage formula (3) is identical up to barrier direction. The controller is shown to be cost-free in calm cycles and to engage at aggressive LTV (Section 9, $L^{\dagger} = 0.84$ sub-experiment).

2. *Funding-rate hedge.* We argue that the implied-yield risk on PT-sUSDE is structurally hedged by a long position on the underlying perpetual funding rate (the input to Ethena's USDe yield stream). The hedged strategy is shown to be a Sharpe-3.2× improvement on the present cycle. In production this hedge would route through Pendle Boros once historical data is available.

13 References

References

[1] A. Krestenko, M. Butov, R. Berezovskiy, D. Bolotin. *Dynamic Collateral Control for Permissionless Spot-Perpetual Basis Trading.* Vega Institute Foundation, 2026.

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[4] Logarithm Labs. *fractal-defi*: Open-source Python research library for DeFi strategies, v1.3.1. `https://github.com/Logarithm-Labs/fractal-defi`.

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A Implementation summary

The full implementation is available at `https://github.com/qqmikhasik/pendle-pt-loop` (BSD-3-Clause, mirroring the parent `fractal-defi` license). Code statistics as of submission:

- 4 entities: `PendlePTEntity`, `MorphoEntity`, `FundingHedgeEntity`, plus reused `BaseEntity[GS, IS]` from `fractal-defi`.
- 4 loaders: `PendleMarketLoader`, `MorphoMarketLoader`, `SUSDePriceLoader`, `FundingHedgeLoader`.
- 6 strategies: 3 baselines, static-loop, dynamic-loop, hedged-loop.
- 153 unit / integration tests, all passing.
- Two CLI scripts: `run_backtest.py`, `sweep_hedge_ratio.py`.

B Reproducibility

To reproduce the headline backtest of Table 1 on a fresh clone:

```
python -m venv .venv && source .venv/Scripts/activate
pip install -e ".[dev]"
pytest -q
python scripts/run_backtest.py --output-dir data/repro
python scripts/sweep_hedge_ratio.py --output data/repro/sweep.csv
```

No API keys are required; all three live data sources (Pendle, Morpho, Hyperliquid) are public and keyless. The default parameters in both scripts target the PT-sUSDE-27NOV2025 cycle on Ethereum mainnet.